

Topology and Spreading Dynamics of the LastFM Asia Social Network

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Complex Networks

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We study the topology of the LastFM Asia network ($N = 7624$, $E = 27806$) against a Configuration Model null model, finding that clustering, degree correlations, and community structure reflect genuine social organisation beyond the degree sequence. We then apply the lifespan method to SIS dynamics on power-law networks ($\gamma = 3.5$ and $\gamma = 2.5$): for $\gamma = 3.5$ a finite threshold is found ($\lambda_c \approx 0.062$) but scan-coverage limitations prevent reliable exponent extraction; for $\gamma = 2.5$ threshold collapse is confirmed ($1/\nu \approx 0.317$, $R^2 = 0.948$) and the differences with the $\gamma = 3.5$ case are discussed.

I. INTRODUCTION

Complex networks provide a mathematical framework for studying systems in which the interaction topology shapes global behaviour [1]. Two questions guide this work: (i) what are the structural properties of a real social network and how do they compare to a maximally random null model with the same degree sequence? (ii) how does degree-distribution heterogeneity control the epidemic threshold and critical exponents of an SIS spreading process? We address both using the LastFM Asia friendship graph [2] for Part I and synthetic power-law networks for Part II.

II. DATA DESCRIPTION

The LastFM Asia dataset is an undirected, unweighted social network of users of the LastFM music platform in Asian countries, where edges represent mutual friendship links [2]. The network is fully connected; the largest connected component (LCC) retains all $N = 7624$ nodes and $E = 27806$ edges and is used throughout. A visualisation of the LCC is shown in Fig. 1.

The edge list is read into a flat neighbour-list vector V of length $2E$, with a degree vector D and frozen/write-head pointer pairs (P_1, P_2) per node. Self-loops and multi-edges are removed, original labels are remapped to indices from $0, \dots, N - 1$, and the GCC is extracted before any analysis.

III. STRUCTURAL ANALYSIS

A. Global properties and degree distribution

Table I summarises the main topological metrics. The network is sparse with $\langle k \rangle = 7.29$ and $k_{\max} = 216$. The average shortest path length $\langle \ell \rangle = 5.175$ and diameter $d = 13$ confirm the small-world property. The slight positive assortativity ($r = 0.017$) is typical of social networks [3].

Figure 2 shows $P(k)$ and $P_c(k)$ on log-log axes. The distribution is heterogeneous with a heavy tail, indicating

Giant connected component

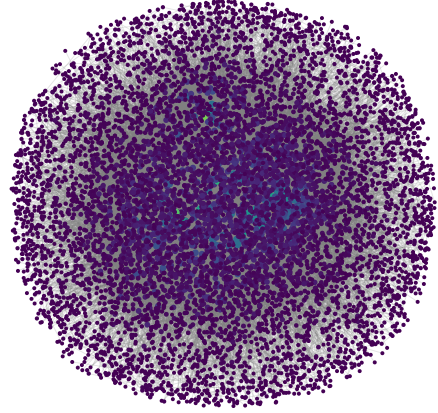


FIG. 1. Visualisation of the LastFM Asia LCC ($N = 7624$, $E = 27806$). Node size is proportional to degree.

TABLE I. Global topological properties of the LastFM Asia LCC.

Property	Value
Nodes N	7624
Edges E	27806
Average degree $\langle k \rangle$	7.294
Maximum degree $k_{\max}$	216
Assortativity r	0.017
Avg. clustering $\langle c \rangle$	0.219
Avg. path length $\langle \ell \rangle$	5.175
Diameter	13
Modularity Q	0.815
Communities (Louvain)	28

the presence of high-degree hubs. The complementary cumulative $P_c(k)$ highlights this behaviour more clearly, with reduced noise at high k .

B. Degree correlations and clustering

Figure 3 shows $k_{nn}(k)$, the average nearest-neighbour degree. Values scatter around and slightly above the

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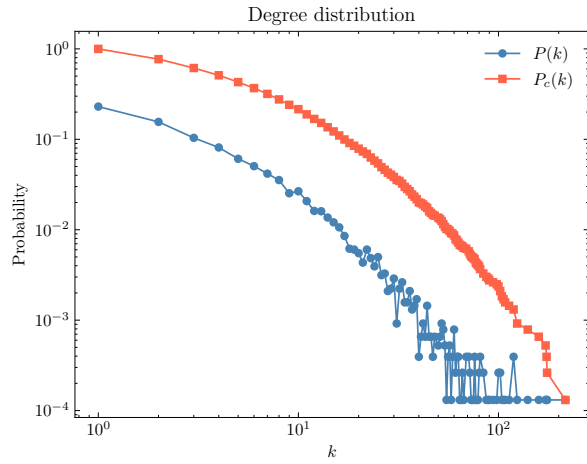


FIG. 2. Degree distribution $P(k)$ (blue) and complementary cumulative distribution $P_c(k)$ (red) of the LastFM Asia LCC on log-log axes.

uncorrelated baseline $\langle k^2 \rangle / \langle k \rangle$ for intermediate degrees, while the highest degree hubs ($k \gtrsim 100$) fall below it. This mixed mixing pattern is consistent with the near-zero assortativity $r = 0.017$ shown in Table I, typical of social networks where high-degree hubs connect to both periphery and intermediate-degree nodes.

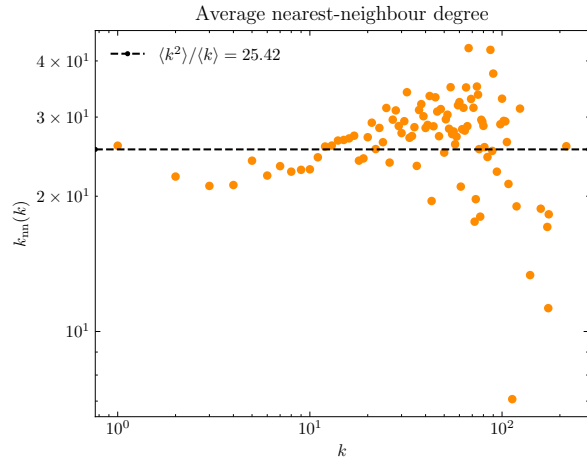


FIG. 3. Average nearest-neighbour degree $k_{nn}(k)$. The dashed line marks the uncorrelated expectation $\langle k^2 \rangle / \langle k \rangle = 25.42$ computed from the real network. Intermediate-degree nodes slightly exceed the baseline while high-degree hubs fall below it, indicating a mixed mixing pattern consistent with the near-zero assortativity $r = 0.017$.

C. Community structure

Community detection is performed using the Louvain algorithm [6], a “greedy” modularity-maximisation heuristic that iteratively reassigns nodes to communities to increase Q , then collapses each community into a single node. The method has no explicit resolution parameter in its standard form, though it is subject to

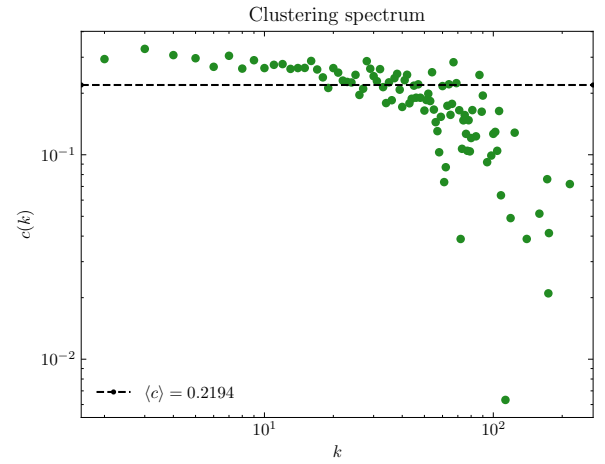


FIG. 4. Degree-dependent clustering coefficient $c(k)$. Strong clustering at low k decreasing toward high k is characteristic of social networks with hierarchical structure.

the “resolution limit”: communities smaller than $\sqrt{E}$ may be merged into larger ones [7]. The algorithm is also non-deterministic; results may vary across runs due to random tie-breaking during node reassignment. Here we report a single run; the high modularity $Q = 0.815$ and large number of communities $n_{\text{comm}} = 28$ are stable across repeated runs on this network.

Louvain yields 28 communities ($Q = 0.815$), well above the CM ensemble mean ($Q = 0.3346 \pm 0.0013$, Table II). Figure 5 shows the network coloured by community; Fig. 6 shows the size distribution, which is heterogeneous with a few large communities and many smaller ones. The CM produces ~ 52 communities at roughly half the modularity, confirming that the observed structure reflects genuine social organisation beyond the degree sequence, not a statistical artefact of $P(k)$.

D. Configuration Model comparison

The CM is generated with stub matching: each node i contributes k_i stubs to a list of length $2E$, which is shuffled uniformly at random; consecutive pairs of stubs are joined into edges, and self-loops and multi-edges are discarded (erased CM). The resulting network preserves the degree sequence exactly in expectation but is otherwise maximally random.

To assess which structural properties are determined by the degree sequence alone, we compare the real network to an ensemble of 100 Configuration Model (CM) realisations preserving $P(k)$. Figures 7, 8, and 9 show $P(k)/P_c(k)$, $k_{nn}(k)$, and $c(k)$ overlaid with the CM. The CM reproduces $P(k)$ by construction but fails to reproduce the degree correlations and the clustering. Table II quantifies this: $\langle c \rangle$, Q , and the number of communities all lie far outside the CM bulk, confirming these are genuine features of the social organisation rather than artefacts of the degree sequence, while assortativity is marginally

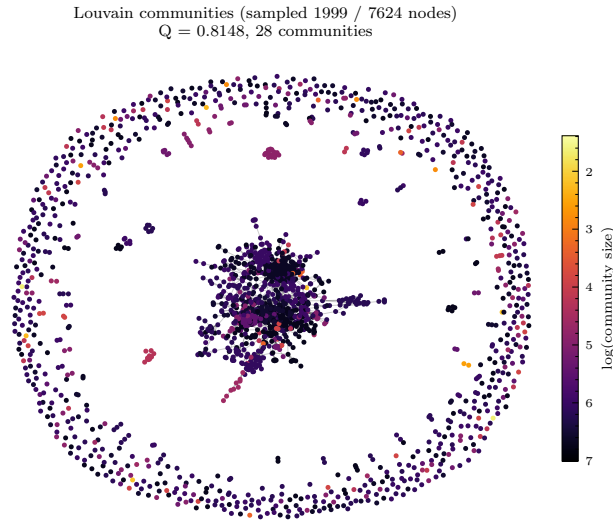


FIG. 5. LastFM Asia LCC coloured by Louvain community (28 communities, $Q = 0.815$).

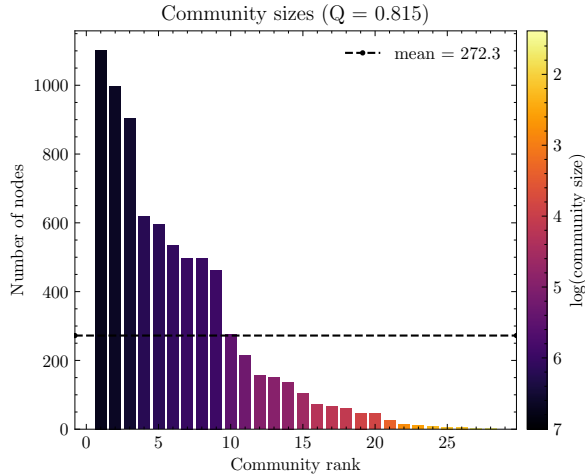


FIG. 6. Community size distribution. A few large communities coexist with many smaller ones.

positive in the real network versus slightly negative in the CM.

IV. SIS DYNAMICS: LIFESPAN METHOD

A. Model and observables

The SIS model runs via the Gillespie algorithm [4]. At each step, a recovery or infection event is drawn with probability proportional to its class rate; the total rate is

$$\lambda_{\text{tot}} = \delta N_I(t) + \lambda E_{\text{act}}(t), \quad (1)$$

where N_I is the number of infected nodes, E_{act} the number of active edges (one infected, one susceptible end-

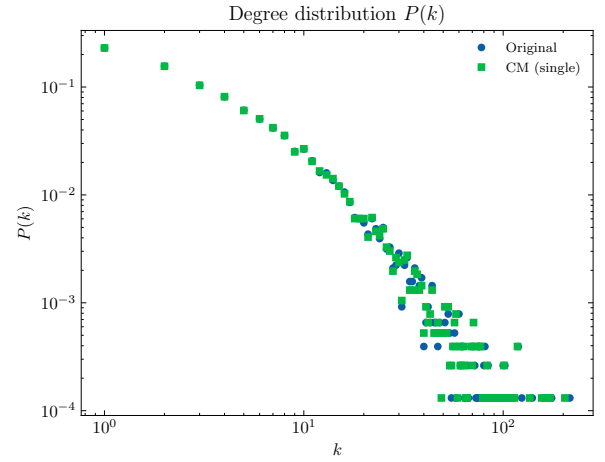


FIG. 7. $P(k)$ for the real network, a single CM realisation, and the CM ensemble mean $\pm 1\sigma$ (100 realisations). The CM preserves $P(k)$ by construction.

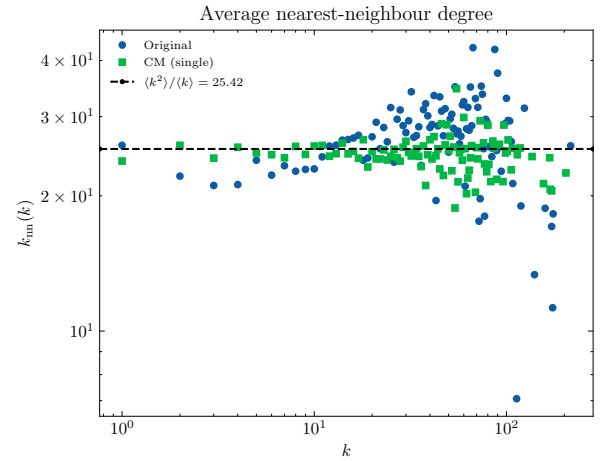


FIG. 8. $k_{\text{nn}}(k)$ for the real network and CM. The dashed line marks the uncorrelated expectation $\langle k^2 \rangle / \langle k \rangle$; the CM recovers it while the real network exceeds it.

TABLE II. Real-network metrics vs. CM ensemble (100 realisations, mean $\pm$ std). Clustering, modularity, and community count lie far outside the CM bulk, confirming they are genuine features of the social organisation rather than consequences of the degree sequence. Assortativity is marginally positive in the real network versus slightly negative in the CM. The ratio $\langle k^2 \rangle / \langle k \rangle$ matches by construction, as both share the same degree sequence.

Metric	Real network	CM ensemble
$\langle c \rangle$	0.219	0.0079 ± 0.0005
Assortativity r	0.017	-0.0096 ± 0.0056
Modularity Q	0.815	0.3346 ± 0.0013
Communities n_{comm}	28	51.82 ± 5.66
$\langle k^2 \rangle / \langle k \rangle$	25.42	24.848 ± 0.058

point), λ the infection rate, and $\delta = 1$. Time advances

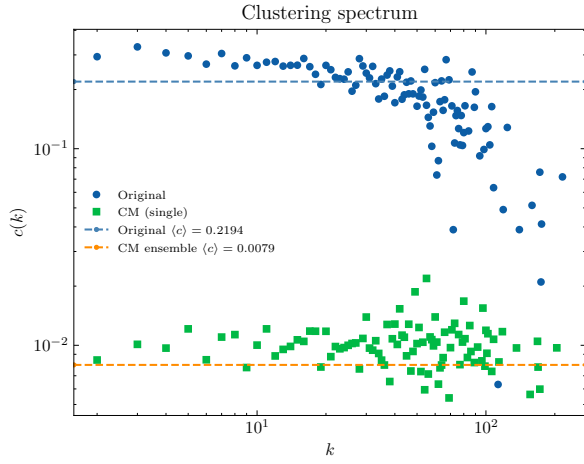


FIG. 9. $c(k)$ for the real network and CM. The CM produces negligible clustering, one order of magnitude below the real value.

as $t \rightarrow t - \lambda_{\text{tot}}^{-1} \ln \xi$, with $\xi \sim U(0, 1)$.

We apply the lifespan method [5]: each realisation starts from a single infected node chosen at random among nodes of minimum degree $k_{\min} = 4$, and is classified as endemic if its coverage exceeds 50% of the network. We measure the order parameter $P_{\text{end}}(\lambda, N)$ and the average lifespan of non-endemic outbreaks $\langle \tau(\lambda, N) \rangle$. Simulations are run on CM power-law networks with $P(k) \propto k^{-\gamma}$, $k_{\min} = 4$, for $\gamma \in \{3.5, 2.5\}$, at seven sizes from $N = 10^4$ to 10^6 .

For infinite systems, close to the critical point, $P_{\text{end}} \propto (\lambda - \lambda_c)^\beta$ and $\langle \tau \rangle \propto |\lambda - \lambda_c|^{-\gamma_1}$. For finite N , the FSS relations are

$$P_{\text{end}}(\lambda, N) = N^{-\beta/\nu} f[(\lambda - \lambda_c)N^{1/\nu}], \quad (2)$$

$$\langle \tau(\lambda, N) \rangle = N^{\gamma_1/\nu} G[(\lambda - \lambda_c)N^{1/\nu}]. \quad (3)$$

The peak position of $\langle \tau \rangle$ scales as

$$\lambda_p(N) = \lambda_c + a N^{-1/\nu}, \quad (4)$$

giving independent estimates of λ_c and $1/\nu$.

B. Peak positions and λ_c , $1/\nu$

1. $\gamma = 3.5$

With both $\langle k \rangle$ and $\langle k^2 \rangle$ finite, HMF theory predicts $\lambda_c = \langle k \rangle / \langle k^2 \rangle > 0$. Figure 10 shows $\langle \tau(\lambda, N) \rangle$: the peaks are noisy and the convergence toward λ_c is slow. The $N = 30000$ spike at $\lambda \approx 0.10$ (lifespan ≈ 2.09 , versus ≈ 1.7 for neighbouring sizes) is a sampling artefact and is filtered before fitting.

Fitting Eq. (4) gives $\lambda_c \approx 0.062$ and $1/\nu \approx 1.281$ ($R^2 = 0.337$). The poor fit reflects a problem of scan coverage: for $N \geq 3 \times 10^5$ the simulated λ -range does not reach below λ_c , so the peak detector takes a value near

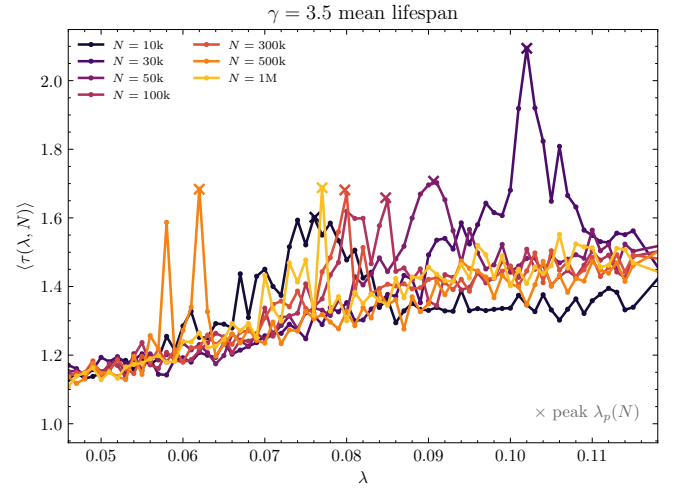


FIG. 10. $\langle \tau(\lambda, N) \rangle$ vs. λ for $\gamma = 3.5$. The peak shifts toward λ_c as N grows. The spike at $N = 30000$ is a statistical artefact.

the lower grid boundary rather than the true maximum. The diagnostic (Fig. 11) confirms this: $\lambda_p^{(\tau)}(N)$ and the onset of $P_{\text{end}} > 0$ lie above the $P_{\text{end}} = 0.005$ contour throughout, meaning the endemic fraction is negligible across the entire scanned range.

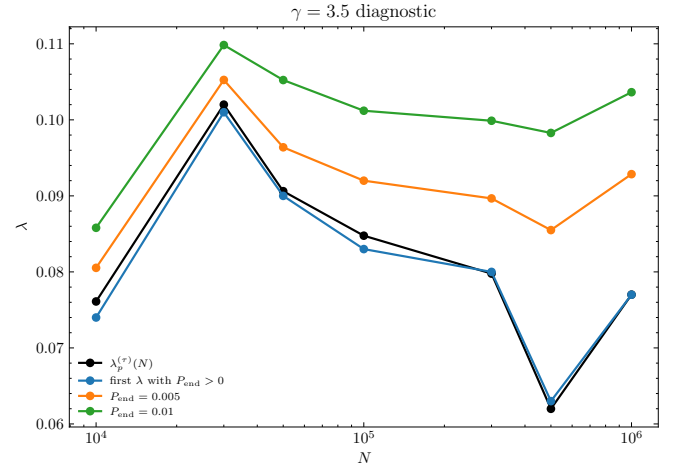


FIG. 11. Diagnostic for $\gamma = 3.5$: peak position $\lambda_p^{(\tau)}(N)$, onset of $P_{\text{end}} > 0$, and contours $P_{\text{end}} = 0.005, 0.01$ vs. N . For $N \geq 3 \times 10^5$ the scan does not cover $\lambda < \lambda_c$.

2. $\gamma = 2.5$

For $\gamma \in (2, 3)$, $\langle k^2 \rangle \sim N^{(3-\gamma)/(\gamma-1)} = N^{1/3}$ diverges with system size, driving $\lambda_c \rightarrow 0$ in the thermodynamic limit. The peak of $\langle \tau \rangle$ shifts monotonically from $\lambda_p \approx 0.019$ at $N = 10^4$ to $\lambda_p \approx 0.005$ at $N = 5 \times 10^5$ (Fig. 12). $N = 10^6$ is excluded from the fit due to anomalous behaviour at very small λ . Fitting with $\lambda_c = 0$ gives $1/\nu \approx 0.317$ ($R^2 = 0.948$), the most reliable result of this section.

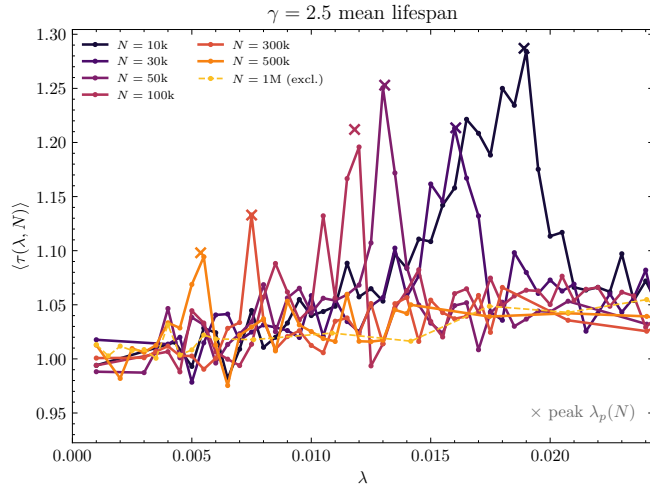


FIG. 12. $\langle \tau(\lambda, N) \rangle$ vs. λ for $\gamma = 2.5$. The peak shifts monotonically toward zero as N grows. $N = 10^6$ (dashed) is excluded from the fit.

C. Exponents γ_1/ν and β/ν

Figure 13 shows the FSS analysis for $\gamma = 3.5$. The peak height gives $\gamma_1/\nu \approx 0.009$ ($R^2 = 0.451$), consistent with a nearly constant lifespan at criticality. The exponent β/ν cannot be extracted: computing $P_{\text{end}}(\lambda_c, N)$ requires interpolation at $\lambda_c \approx 0.062$, which lies outside the scanned range for the three largest sizes (confirmed by Fig. 11).

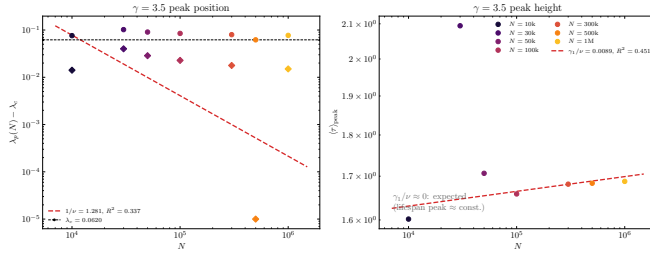


FIG. 13. FSS for $\gamma = 3.5$: $\lambda_p(N) - \lambda_c$ (left) and $\langle \tau \rangle_{\text{peak}}$ (right) vs. N on log-log axes ($R^2 = 0.337$ and 0.451).

For $\gamma = 2.5$ (Fig. 14), the peak height decreases with N , giving $\gamma_1/\nu \approx -0.039$ ($R^2 = 0.884$). The negative sign is anomalous from a standard FSS perspective. Since $\lambda_c = 0$, $P_{\text{end}}(0, N) = 0$ identically for all N , so β/ν is inapplicable by construction: there is no finite λ_c at which to evaluate the order parameter.

D. Data collapse

The attempted collapse of $\langle \tau \rangle$ vs. $(\lambda - \lambda_c)N^{1/\nu}$ for $\gamma = 3.5$ is shown in Fig. 15. The curves do not collapse: the large estimated $1/\nu \approx 1.281$ amplifies even small deviations $\lambda - \lambda_c$ for large N , stretching those curves far to the right while the smaller sizes remain compressed near the origin. This directly reflects the unreliable λ_c and

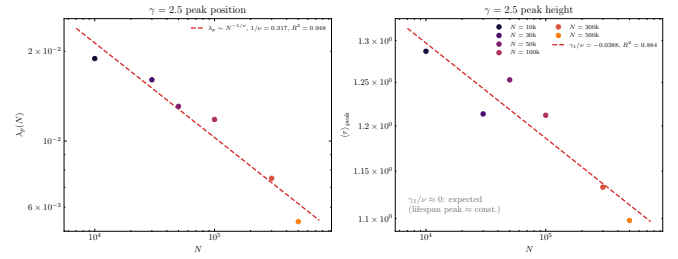


FIG. 14. FSS for $\gamma = 2.5$: $\lambda_p(N)$ with fit ($1/\nu = 0.317$, $R^2 = 0.948$, left) and $\langle \tau \rangle_{\text{peak}}$ ($\gamma_1/\nu = -0.039$, right).

$1/\nu$ estimates caused by the scan-coverage gap.

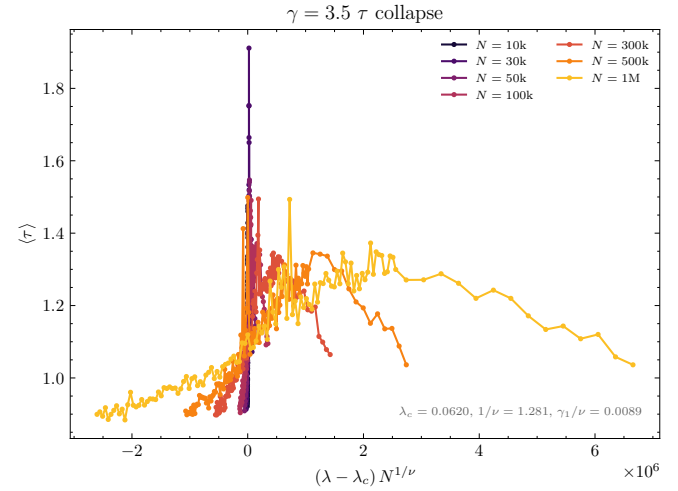


FIG. 15. Attempted FSS collapse for $\gamma = 3.5$: $\langle \tau \rangle$ vs. $(\lambda - \lambda_c)N^{1/\nu}$ ($\lambda_c = 0.062$, $1/\nu = 1.281$).

For $\gamma = 2.5$ (Fig. 17 and diagnostic Fig. 16), the collapse of $\langle \tau \rangle/N^{\gamma_1/\nu}$ also fails: because $\gamma_1/\nu < 0$, dividing by $N^{\gamma_1/\nu}$ increases the rescaled lifespan for large N , separating the curves rather than aligning them. The diagnostic confirms that the peak and the P_{end} onset both converge to zero as expected. The scan coverage is adequate, so the collapse failure is because of the anomalous exponent.

E. Validation of the method

The lifespan method was designed for systems with a well-defined finite critical point [5]. For $\gamma = 3.5$, the method is in principle applicable, but the practical obstacle is technical: the λ scan must bracket λ_c for every system size, which our simulations fail to do at large N . With adequate scan coverage, a clean FSS analysis should be achievable.

For $\gamma = 2.5$, the situation is fundamentally different. The threshold $\lambda_c \rightarrow 0$ in the thermodynamic limit because $\langle k^2 \rangle$ diverges, which places the critical point exactly at $\lambda = 0$ for an infinite system. This has three consequences. First, the peak position fit (Eq. 4) still works

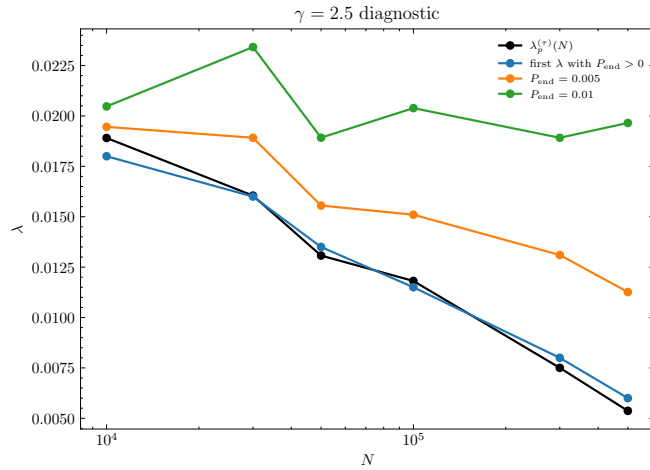


FIG. 16. Diagnostic for $\gamma = 2.5$: peak position, P_{end} onset, and contours vs. N . All quantities converge toward zero, confirming the scan brackets λ_c for all sizes.

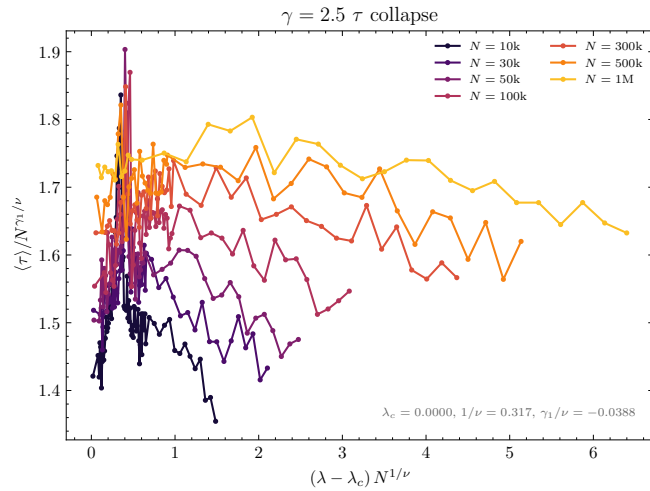


FIG. 17. Attempted FSS collapse for $\gamma = 2.5$: $\langle \tau \rangle / N^{\gamma_1/\nu}$ vs. $(\lambda - \lambda_c)N^{1/\nu}$ ($\lambda_c = 0$, $1/\nu = 0.317$).

and gives a well-defined $1/\nu$ because the peak approaches zero as a power law of N , not because it converges to a finite λ_c . Second, the order parameter exponent β/ν is inaccessible: $P_{\text{end}}(0, N) = 0$ trivially, and there is no finite λ_c at which to evaluate the scaling relation. Third, the exponent γ_1/ν is anomalous (< 0): in the $\gamma < 3$ regime, larger networks contain higher-degree hubs that quench non-endemic outbreaks more efficiently, so the peak lifespan decreases with N rather than growing. This violates the standard FSS assumption that $G[0] = \text{const} > 0$ is finite and positive, and prevents a meaningful data collapse. Logarithmic corrections and strong hub-driven finite-size effects are expected throughout this regime [5].

V. CONCLUSIONS

The structural analysis shows that clustering, degree correlations, and community structure in the LastFM Asia network lie far outside the CM ensemble, confirming they reflect higher-order social organisation beyond the degree sequence.

For the SIS model, the lifespan method correctly identifies the qualitative behaviour in both regimes. For $\gamma = 3.5$, peak positions point to a finite threshold ($\lambda_c \approx 0.062$), but the scan does not bracket λ_c for the three largest sizes, making the exponents unreliable ($1/\nu \approx 1.281$, $R^2 = 0.337$). The exponent β/ν cannot be extracted and no data collapse is achieved. For $\gamma = 2.5$, threshold collapse is robustly confirmed ($1/\nu \approx 0.317$, $R^2 = 0.948$); however, $\gamma_1/\nu \approx -0.039$ is anomalous and β/ν is inapplicable by construction ($\lambda_c = 0$), so the method only partially works in this regime. Both cases would benefit from denser λ -scans below the current grid boundaries and more realisations per system size.

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